Draft version April 4, 2026
Typeset using LaTeX **preprint** style in AASTeX7.0.1

Faint CO Absorption Line Census in Active Galaxies from the ALMA Archive

Hammad Abdullah[1] and Benjamin D. Boizelle[2]

[1]*Department of Physics and Astronomy, Rutgers University, 136 Frelinghuysen Rd, Piscataway, NJ 08854, USA*
[2]*Department of Physics and Astronomy, N284 ESC, Brigham Young University, Provo, UT 84602, USA*

ABSTRACT

When an active galaxy is observed at mm/sub-mm wavelengths, the low quantized rotational states (J) of CO can reveal emission and absorption from line-of-sight clouds in its spectra. Spectral analysis of the absorption features from clouds in front of the host galaxie's active galactic nucleus (AGN) reveal their kinematic and properties. We present a census of faint CO absorption lines in archival ALMA observations of 20 early-type galaxies at $z < 0.06$. An emphasis is placed on identifying narrow, optically thin features that may have been overlooked in prior studies. Out of the 20 targets, 10 show CO absorption in at least one data set. The detections reveal a range of cloud populations, including faint, narrow absorbers consistent with diffuse inflowing gas, broader optically thick features associated with more chaotic environments, and clouds whose motions are consistent with disk-like rotation. High spectral resolution proved essential for resolving narrow absorption features that would otherwise have been averaged out with adjacent channels in more coarsely binned observations. Comparisons with previous studies suggest that a previously undetected population of low-velocity-dispersion ($\sigma_{\rm v} < 10$ km s^{-1}), optically thin, inflowing clouds exists. These results support the view that cold molecular gas near AGN is diverse in both kinematics and optical depth, and that faint absorption-line studies can provide meaningful constraints on AGN accretion.

1. INTRODUCTION

Black holes form when a gas cloud or star collapses in on itself and the inward force of gravity is strong enough to prevent light from escaping. They can be split into three categories based on their mass: stellar-remnant, intermediate, and supermassive. Here, we focus on supermassive black holes (SMBHs) that are found at the centers of galaxies. A fraction of these are observed to be actively accreting material and are thus denoted as active galactic nuclei (AGN). These AGN are among the universe's most powerful, brightest, and mysterious phenomena, as these compact objects can launch jets of plasma detected far beyond the host galaxy's confines. They are fueled by gas and stars spiraling toward the central gravitational potential, the most active of which may consume several solar masses each year ($M_\odot$ yr^{-1}; Salpeter 1964). The subsequent radiation from this accretion process may create feedback in the form of high-energy emitted radiation or galactic winds that push back against inflowing gas within the central regions of the host galaxy (Schawinski et al. 2007). Depending on the accretion rate, the AGN of one galaxy may output greater radiation and feedback than another.

The mechanisms that fuel AGN remain subjects of active discussion, especially how gas at large distances funnels inward toward the centers of massive galaxies. One proposed mechanism is cold-chaotic accretion (CCA), a process in which cold molecular gas clouds composed primarily of molecular hydrogen (H_2) funnel toward the galactic nucleus through stochastic collisions between adjacent clouds (Gaspari et al. 2013). These clouds, with kinetic temperatures ranging between 5 K and 40 K, are frequently seen in mm/sub-mm observations. Calculating the amount of H_2 within a molecular cloud approaching or receding from an SMBH would provide insight into AGN fueling and feedback. The column density of H_2 ($N_{\mathrm{H_2}}$; molecules per square cm) is a measure of the total material along a particular line of sight (LOS) and is often inferred for gas seen in front of a background AGN.

Unfortunately, $N_{\mathrm{H_2}}$ cannot be measured directly at mm/sub-mm wavelengths. The next most common molecule, carbon monoxide (CO), is typically used as a tracer for H_2. Within cold gas clouds, inelastic collisions populate quantized rotational levels of CO and other molecules. Akin to electron transitions in an atom, the rotational energy state J can change by the emission or absorption of a photon at specific frequencies. Examples of these CO lines are CO(1–0) (transition from $J = 1$ to the ground state), CO(2–1) (transition from $J = 2$ to $J = 1$), and CO(3–2) (transition from $J = 3$ to $J = 2$). CO absorption seen against bright continuum sources such as AGN arises from transitions between J and $J + 1$ states. Measurements of CO column density (N_{CO}) can thus be obtained through the spectral analysis of these emission or absorption lines (Mangum and Shirley 2015). A calibration is then used to convert N_{CO} into $N_{\mathrm{H_2}}$, with a typical ratio of 1:10,000 that is highly dependent on the gas environment.

The observed central frequency of a particular CO rotational transition, modeled with a Gaussian function, can be converted into a centroid velocity (v_0) relative to the systemic radial motion of the galaxy (v_{sys}). The relative centroid velocity, $v_{-}v_{\mathrm{sys}}$, can provide critical insight into the origin of the gas cloud. Due to the nature of galaxy dynamics, this results in three likely regimes in which a CO spectral feature can be observed. Absorption lines with $|v - v_{\mathrm{sys}}| \lesssim 50\,\mathrm{km\,s^{-1}}$ likely correspond to gas from a highly inclined rotating disk in circular motion around the central SMBH. At these locations, the relative motion is mostly perpendicular to our line of sight. Redshifted lines with positive velocity, $v_0 - v_{\mathrm{sys}} \gtrsim 50\ \mathrm{km\,s^{-1}}$, indicate gas inflowing toward the central AGN. However, the full cloud velocity or distance from the AGN cannot be inferred from these data alone. The final regime is blueshifted absorption, with $v_0 - v_{\mathrm{sys}} \lesssim -50\ \mathrm{km\,s^{-1}}$, which represents gas outflows from the galaxy's central region. The spread of each line profile, or standard deviation from v_0, is denoted as velocity dispersion (σ_{v}) and is measured in $\mathrm{km\,s^{-1}}$. Broader features with $\sigma_{\mathrm{v}} > 10\,\mathrm{km\,s^{-1}}$ reflect molecular clouds with larger internal velocity variations stemming from chaotic environments, such as near the central SMBH region, lines of sight through multiple clouds with different intrinsic velocities, or large self-gravitating clouds. Narrower features highlight clouds with more uniform internal dynamics. These clouds may be farther from the hot central SMBH region and within a more stable, less turbulent environment, or they may be part of the outer, more stable regions of a larger, more chaotic cloud. Analysis of CO rotational lines therefore provides not only the amount of molecular gas within the line of sight, but also its overall dynamics.

A census of mm/sub-mm AGN continua that measures the prevalence and characteristics of cold molecular clouds would help constrain accretion models such as CCA. Performing this type of analysis requires a powerful radio telescope array capable of providing data with high angular and spectral resolution. Fortunately, one such array, the Atacama Large Millimeter/Submillimeter Array (ALMA), was put into operation in 2012 and has been imaging nearby active galaxies in the mm/sub-mm regime ever since. As a result, a plethora of archival observations of galaxy targets with potential CO spectral features remain open for analysis. One recent study by Rose et al. (2024) characterized CO absorption in 12 nearby ($z < 0.06$) active galaxies using archival ALMA imaging to examine the cold-cloud population. In some cases, the CO absorption was embedded within emission features, in part due to large spectral extraction regions. After isolating the absorption lines, they fitted Gaussian functions to eight of the 12 target galaxies. The study found the presence of two distinct gas populations seen in front of the AGN: one with $\sigma_{\rm v}$ between 10 and 100 $\rm km\,s^{-1}$ and a highly positive $v_0 - v_{\rm sys}$, indicating net movement toward the central SMBH, and another with $\sigma_{\rm v} < 10\,{\rm km\,s^{-1}}$ and $v_0 - v_{\rm sys}$ close to zero, consistent with a rotating disk.

It is worth noting that the Rose et al. analysis focused on detecting the most prominent (deep and/or broad) features, leaving potential narrow and faint lines uncharacterized. These fainter features may trace diffuse, optically thin, and stable gas compared to the more chaotic, high-$\sigma_{\rm v}$ gas they identified. One such case explored by Boizelle et al. (2025) detected a faint CO line in multiple transitions and connected it to a foreground filament close to the galactic nucleus.

In this analysis, we search for new CO absorption lines in 20 nearby ($z \leq 0.06$) elliptical and lenticular galaxies. Most targets host a radio-loud AGN, which aids in identifying faint absorption lines. In addition to the sample of nearby galaxies, we also identified ALMA observations of high-redshift quasar targets whose spectral setups allow for CO detection. Due to their bright continuum emission, these were used as calibrators for science observations.

The thesis is organized as follows. In Section 2, I detail the physical properties of the target galaxies in Table 1, as well as their imaging observations in Table 2, and I describe the line-fitting model and Monte Carlo resampling procedure. In Section 3, I showcase select spectra from four non-detection cases, as well as four detection cases, to highlight the complexity and nuance involved in analyzing spectral lines for CO absorption. In Section 4, I offer an overall analysis of this census and its implications for future absorption-line studies and CO surveys. Additional information presenting the remaining spectra, as well as one from the high-redshift quasar calibrator target, is provided in the Appendix.

2. METHODOLOGY

2.1. *Galaxy Targets*

The sample of galaxies was chosen based on their morphologies and peak continuum brightness ($I_{\nu,\rm peak}$) measured in millijanskys per beam ($\rm mJy\,beam^{-1}$). The morphological description of the sample is as follows: 11 elliptical (E), 4 elliptical-lenticular (E–S0), 4 lenticular (S0), and one spiral (S). Since these targets are nearby, within $z \approx 0.06$, they are all classified as early-type galaxies that are in the later stages of their evolution. Locations, redshifts, and inclination angles can all be found in Table 1. Information detailing the imaging of these targets, the corresponding CO transitions, and their spectral properties is listed in Table 2.

Table 1. Target Galaxy Properties

Source	RA, Dec (J2000)	$z_{\rm helio}$	Redshift reference	Inclination (deg)	Morphology
3C 75	02:57:42, +06:01:34	0.023153	de Vaucouleurs et al. (1991)	0	E
ESO 208–G021	07:33:56, −50:26:35	0.003619	Ogando et al. (2008)	55	S0
Hydra A	09:18:06, −12:05:44	0.054878	Smith et al. (2004)	79	E−S0
IC 1459	22:54:11, −36:27:43	0.006011	Wong et al. (2006)	74	E
IC 4296	13:36:39, −33:57:57	0.012465	Allison et al. (2014)	90	E
J1427−4206	14:27:56, −42:06:19	1.522	White et al. (1988)	···	Quasar
NGC 0547	12:15:49, −16:31:08	0.021115	de Vaucouleurs et al. (1991)	90	E
NGC 1052	02:41:05, −08:15:21	0.005204	Koss et al. (2022)	70	E
NGC 1277	03:19:51, +41:34:24	0.016749	SDSS-IV DR13	52	S0
NGC 1380	03:26:28, −34:58:34	0.006264	Iodice et al. (2019)	90	S0
NGC 2110	05:52:11, −07:27:22	0.007849	Koss et al. (2022)	46	E−S0
NGC 3078	09:58:25, −26:55:36	0.008606	Allison et al. (2014)	51	E
NGC 315	00:57:49, +00:21:09	0.016485	Trager et al. (2000)	57	E
NGC 3557	11:09:58, −37:32:21	0.010270	Lagattuta et al. (2017)	56	E
NGC 383	01:07:25, +32:24:45	0.017005	Smith et al. (2000)	40	E−S0
NGC 3862	11:45:05, +19:36:23	0.021595	de Vaucouleurs et al. (1991)	0	E
NGC 4261	12:19:23, +05:49:30	0.007261	SDSS-IV DR13	56	E
NGC 4374	12:25:04, +12:53:13	0.003392	Cappellari et al. (2011)	40	E
NGC 5084	13:20:17, −21:49:39	0.005741	Koribalski et al. (2004)	90	S0
NGC 6328	17:23:41, −65:00:37	0.014428	Beuchert et al. (2018)	15	Sa
NGC 6861	20:07:19, −48:22:12	0.009437	Ogando et al. (2008)	51	E−S0

Note— Right ascension, declination, and redshifts were obtained from the NASA/IPAC Extragalactic Database (NED). Morphologies and estimated inclination angles were obtained from the HyperLeda database.

Table 2. ALMA Imaging Properties

Source	Project code	Obs. date (yyyy-mm-dd)	Ang. res. (arcsec)	Int. time (s)	Line(s) CO	$I_{\nu,\rm peak}$ (mJy beam^{-1})	Δv (km s^{-1})	$\sigma_{\rm noise}$ (mJy beam^{-1})
3C 75	2017.1.01638.S	2018-09-21	0.18	2400	(3–2)	26	13.89	0.240
					(3–2)	21		0.263
ESO 208–G21	2017.1.00301.S	2018-01-15	0.24	1900	(2–1)	23	5.10	0.669
Hydra A	2016.1.01214.S	2016-10-23	0.19	2700	(2–1)	70	2.01	1.003
IC 1459	2015.1.01572.S	2016-04-11	0.73	700	(2–1)	298	1.28	2.438
	2017.1.01638.S	2018-09-13	0.20	2900	(3–2)	244	13.65	0.369
IC 4296	2015.1.01572.S	2014-07-23	0.46	3100	(2–1)	213	1.29	1.354
J1427−4206	2018.1.00490.S	2018-10-24	0.64	300	(2–1)	3719	3.20	4.309
	2021.1.01195.S	2022-01-28	1.02	6800	(2–1)	2876	3.20	2.236
	2022.1.01694.S	2023-03-05	1.10	8700	(2–1)	6024	0.80	4.599
NGC 0547	2017.1.00301.S	2018-01-18	0.26	5400	(2–1)	79	5.17	0.425
	2013.1.01225.S	2015-08-08	0.13	4100	(3–2)	334	13.61	2.68
	2015.1.00591.S	2016-05-26	0.48	1200	(1–0)	741	40.84	1.977
NGC 1052								

Table 2 *continued on next page*

Table 2 *(continued)*

Source	Project code	Obs. date (yyyy-mm-dd)	Ang. res. (arcsec)	Int. time (s)	Line(s) CO	$I_{\nu,\mathrm{peak}}$ (mJy beam^{-1})	Δv (km s^{-1})	σ_{noise} (mJy beam^{-1})
	2015.1.01290.S	2015-10-23	0.03	620	(2–1)	429	20.42	0.944
	2015.1.01290.S	2017-09-06	0.01	3500	(3–2)	326	13.61	2.152
NGC 1277	2024.1.00779.S	2025-05-10	0.20	4900	(2–1)	11	5.17	0.684
NGC 1380	2013.1.00229.S	2015-06-11	0.15	2700	(2–1)	5	2.56	0.491
	2015.1.00497.S	2016-01-11	2.10	230	(1–0)	5	2.56	0.867
NGC 2110	2012.1.00474.S	2015-05-14	0.59	2200	(2–1)	4	1.28	0.468
	2015.1.00419.S	2016-06-22	0.50	580	(2–1)	40	0.64	3.086
	2021.1.00812.S	2021-11-05	0.21	760	(1–0)	4	5.12	0.621
NGC 3078	2015.1.00878.S	2016-07-21	0.31	1200	(2–1)	90	5.12	0.940
NGC 315	2017.1.00301.S	2018-09-19	0.18	5100	(2–1)	215	5.16	0.714
	2019.1.00036.S	2021-07-01	0.23	1800	(2–1)	219	1.29	1.128
	2022.1.01040.S	2023-09-21	0.06	4200	(2–1)	284	1.29	1.119
NGC 3557	2015.1.00591.S	2015-12-26	2.30	610	(1–0)	51	2.57	5.430
NGC 383	2015.1.00419.S	2016-06-21	0.57	120	(2–1)	88	0.65	6.995
	2016.1.00437.S	2017-08-16	0.12	1700	(2–1)	19	0.65	1.108
	2019.1.00582.S	2021-09-05	0.02	7800	(2–1)	77	1.29	0.497
NGC 3862	2015.1.00598.S	2016-06-11	0.54	640	(2–1)	109	1.30	4.926
	2018.1.00397.S	2019-08-27	0.11	2300	(2–1)	64	1.30	0.897
NGC 4261	2017.1.00301.S	2018-01-19	0.26	1900	(2–1)	252	5.12	1.029
	2017.1.01638.S	2018-01-21	0.18	2300	(3–2)	227	13.80	1.203
	2023.1.00842.S	2023-12-03	0.08	6600	(3–2)	177	1.71	1.257
NGC 4374	2013.1.00828.S	2015-08-16	0.23	3000	(2–1)	134	1.28	2.890
NGC 5084	2015.1.00598.S	2016-04-30	0.51	540	(2–1)	58	0.64	2.962
NGC 6328	2015.1.01359.S	2016-09-03	0.19	6800	(2–1)	309	10.31	0.529
	2017.1.01638.S	2018-01-21	0.20	2700	(3–2)	226	13.77	0.493
NGC 6861	2013.1.00229.S	2014-09-01	0.26	1400	(2–1)	27	2.56	2.20
	2016.1.01135.S	2016-11-01	0.19	3000	(3–2)	18	13.67	0.529

Note—ALMA imaging information for each data set organized by galaxy name and then project code. Angular resolutions are rounded to two decimal places. Integration times are rounded to two significant figures. Continuum peak intensities are rounded to the nearest integer. Δv is the channel binning. σ_{noise} is the observed uncertainty from the line-free channels. J1427−4206 is not part of the main sample.

2.2. *Obtaining Spectra and Analysis Procedure*

2.2.1. *Cleaning Data*

Dr. Boizelle and I downloaded the raw imaging data (project codes found in Table 2) from the ALMA archive. After standard pipeline calibrations using the Common Astronomy Software Applications (CASA; McMullin 2007), we performed imaging and self-calibration to produce the spectra used in this analysis. I had performed this procedure individually for the data sets of NGC 1052 and NGC 315. The `tclean` task with a *specmode* of *mfs* was used to image the continuum, and the *gaincal* task improved the atmospheric phase solutions and final signal-to-noise (S/N) of a given measurement set. Some errant flux measurements seen in the amplitude-versus-channel plots generated by the `plotms` task were masked. For *tclean*, the following parameters were used: Hogbom deconvolver and Briggs weighting (robust parameter of 0.5). Subsequent phase self-calibrations with a reference antenna holding the least percentage of flagged data and a central array location were applied until the S/N either increased slightly or decreased from the pre-calibrated set. The measurement set with the highest S/N at the end of self-calibration was chosen for spectral cube construction. `tclean` was used again with *specmode* now set to *cube* and the same deconvolver and weighting, along with a barycentric reference frame. After applying the primary beam correction output, we created a subsequent data cube file containing the relevant spectral information for each observation.

2.2.2. *Model Fitting*

From the final data cube file, I extracted the spectra for each AGN at its peak-intensity pixel. This helps to minimize contamination by CO emission, which tends to be extended.

I converted channels into frequencies by

$$\nu_{\rm obs} = \frac{\nu_{\rm ref} + (i - i_{\rm ref})\,\Delta\nu}{10^9} \tag{1}$$

where $\nu_{\rm obs}$ is the observed frequency in GHz, i is the channel number, $i_{\rm ref}$ is the reference channel number, and $\Delta\nu$ is the individual frequency width. This frequency array is then converted into velocities:

$$z_{\rm obs} = \frac{\nu_{\rm rest}}{\nu_{\rm obs}} - 1 \qquad z_{\rm pec} = \frac{1 + z_{\rm obs}}{1 + z_{\rm sys}} - 1 \qquad 1 + z_{\rm pec} \equiv \beta_z \qquad v - v_{\rm sys} = c\left(\frac{\beta_z^2 - 1}{\beta_z^2 + 1}\right) \tag{2}$$

Here, $\nu_{\rm rest}$ is the rest frequency of the relevant molecular rotational transition, $z_{\rm obs}$ is the observed redshift, and $z_{\rm pec}$ is the redshift of gas with respect to the systemic motion of the observed galaxy ($z_{\rm sys}$ and $v_{\rm sys}$). These velocities can then be used to construct an Intensity-versus-Velocity plot of the observed spectra.

I then normalized these spectra to the background continuum source using a low-order polynomial fit on the line-free regions. In most cases, a degree-one linear fit works well to represent the intensity of the AGN. However, a higher-order fit can be necessary to better capture the continuum. This normalized continuum spectrum can be expressed in either velocity, frequency, or channel space.

Next, I applied a multi-Gaussian fitting model within frequency space to capture the absorption shape, if present. The `scipy.optimize.least_squares` method, with an initial guess and bounds, was used to constrain the best-fit parameters for the model described by Equation 3:

$$I(\nu) = m + b\nu + \sum_{i=1}^{n}(-A_i)\exp\left(-\frac{(\nu - \nu_{0,i})^2}{2\sigma_{\nu,i}^2}\right) \tag{3}$$

$$\sigma_{\rm v} = \frac{c\sigma_\nu}{\nu_{\rm rest}} \tag{4}$$

where $m + b\nu$ is the continuum, $\nu_{0,i}$ is the frequency centroid, $\sigma_{\nu,i}$ is the frequency dispersion, and A_i is the amplitude profile. The frequency centroids can be converted into velocity centroids via Equation 2, while σ_ν can be converted to $\sigma_{\rm v}$ via Equation 4. If emission is present within the spectra, the same fitting model is used, but with a positive amplitude term. The emission model is then subtracted from the spectrum. The imaging in two cases, NGC 2110 (2012.1.00474.S, 2021.1.00812.S) and NGC 1380 (2015.1.00419.S), had the target move drastically between observation channels. In such cases, taking the peak-intensity pixel would be meaningless, as in some channels the target would be there, while in others it would be observed somewhere else. To account for this, the spectra for these three data sets were obtained by taking an elliptical region of the *.fits* file centered at an appropriate pixel that best captures the target spectra. The semi-major and semi-minor axes for the ellipse are obtained from the associated *.fits* file.

2.2.3. *Error Analysis*

After obtaining the best-fit parameters, I used a Monte Carlo resampling procedure to generate error bounds. This is done by adding random noise to a synthetic spectrum matching the model fit before refitting. This is done many times, with noise values drawn from a normal distribution with standard deviation matching that of the line-free regions of the normalized continuum spectra. To account for uncertainty in the absorber centroid location due to channel binning, I also shifted $\nu_{\rm ref}$ in Equation 1 by a random amount based on a normal distribution centered on a quarter of the frequency binning. To keep the order of Gaussian terms identical when there are multiple absorbers present within a spectrum, initial guesses match the best parameters of the model fit. Constant and linear terms are kept at 1.00 and 0.00, respectively, since the resampling procedure is performed on a model fit normalized to a continuum. After 10,000 iterations, I took the 15.9th and 84.1st percentiles of the resulting parameter distribution and subtracted them from the initial model value to yield the 1σ error associated with each Gaussian parameter.

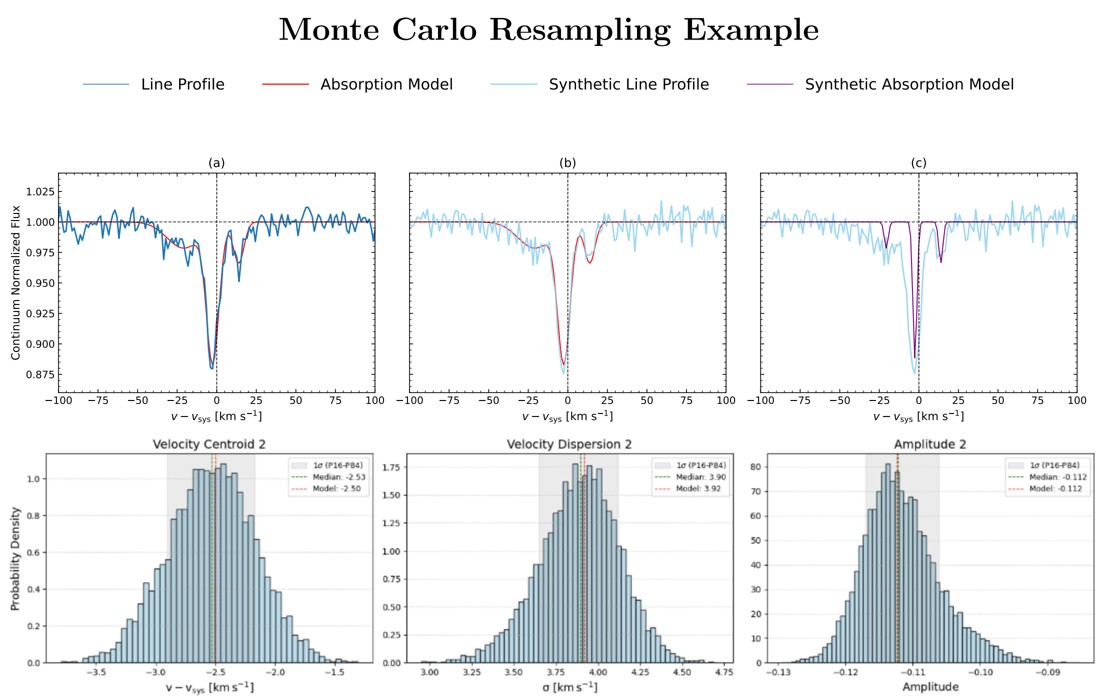


Figure 1. Monte Carlo resampling procedure visualized. The original spectra overlaid with the initial absorption-line fitting model are shown in (a). The synthetic spectra generated from the initial absorption model are shown in (b), while the new absorption-line fitting model for the synthetic spectra is shown in (c). The histograms showcase the Gaussian parameter distribution of 10,000 synthetic spectra for the deepest absorption feature of the original spectra (centroid 2).

2.3. *Calculable Quantities*

2.3.1. *Optical Depth*

The depth of a detected absorption feature can reveal the optical thickness of the molecular gas cloud. I calculated this quantity using the standard formula in Equation 5,

$$\tau_\nu = -\ln\left(\frac{I_v}{I_{\nu,\mathrm{cont}}}\right) \tag{5}$$

where τ_ν is the frequency-dependent optical depth, I_v is the observed spectrum with emission subtracted if present, and $I_{\nu,\mathrm{cont}}$ is the continuum. From this array of optical depths, I then calculate the integrated opacity (τ_{CO}) of the CO line by adding positive τ_ν values associated with an absorption region.

$$\tau_{\mathrm{CO}} = \int \tau_\nu \, dv \tag{6}$$

where dv is the channel binning. The associated error of integrating over N_{chan} channels can be obtained by Equation 7,

$$\sigma_{\tau_{\mathrm{CO}}} = \sigma_\tau \sqrt{N_{\mathrm{chan}}}\, \Delta v \tag{7}$$

where σ_τ is the standard deviation of the absorber-free τ_ν spectrum, and Δv is the average velocity channel width.

When absorption is not detected against an AGN spectrum, an upper limit on τ_{CO} can still be obtained via Equation 7. As an alternative method, an arbitrary feature, usually the deepest, within the normalized continuum spectra and $v - v_{\mathrm{sys}} = \pm 600\,\mathrm{km\,s^{-1}}$, is chosen to represent a limiting extreme that, when integrated over a small N_{chan}, represents the maximum integrated opacity.

2.3.2. *Column Density and Excitation Temperature*

After calculating τ_{CO} for a given CO transition state within a dataset, we then calculate the excitation temperature T_{ex} by (Mangum and Shirley 2015):

$$\frac{\tau_{\mathrm{CO}(u-l)}}{\tau_{\mathrm{CO}(l-ll)}}\left[\frac{\nu_{\mathrm{CO}(u-l)}}{\nu_{\mathrm{CO}(l-ll)}}\right]^3 \frac{g_l A_{ull}}{g_u A_{ul}} = \frac{1 - e^{-h\nu_{\mathrm{CO}(u-l)}/kT_{\mathrm{ex}}^{ul}}}{e^{h\nu_{\mathrm{CO}(l-ll)}/kT_{\mathrm{ex}}^{ul}} - 1} \tag{8}$$

Assuming local thermodynamic equilibrium (LTE) conditions, for a measured (or adopted) T_{ex},

$$N_{\mathrm{CO}} = Q_{\mathrm{rot}}\left(\frac{8\pi\nu_{u-l}^3}{c^3}\right)\frac{g_l}{g_u A_{ul}}\frac{1}{1 - e^{-h\nu_{\mathrm{CO}(u-l)}/kT_{\mathrm{ex}}^{ul}}}\int \tau_{\nu(u-l)}\, dv \tag{9}$$

where u represents the upper rotational level, l the lower rotational level, and ll the lower-lower rotational level. For example, a (2–1) and a (1–0) transition would have $u = 2$, $l = 1$, and $ll = 0$. Here, ν_{CO} are the rest frequencies, $g_l = 2l + 1$ is the rotational degeneracy number, and A_{ul} are the Einstein A coefficients taken from *Splatalogue*. Q_{rot} is the temperature-dependent rotational partition function for CO determined by Mangum and Shirley (2015). Assuming LTE, Equation 9 can be used to calculate N_{CO} for any CO rotational transition state. This value can then be converted into $N_{\mathrm{H_2}}$ through the CO/H_2 abundance ratio of 3.2×10^{-4} (Sofia et al. 2004).

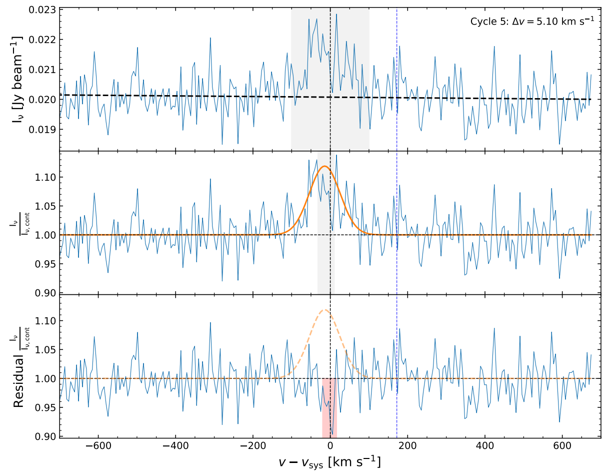
If both CO(2–1) and CO(1–0) transitions have individual spectra for a target galaxy, then Equation 8 can be followed to numerically determine $T_{\rm ex}$ of the absorbing gas by the intersection of the left-hand constant with the exponential term on the right as a function of $T_{\rm ex}$. Asymmetric error bounds for $T_{\rm ex}$ can then be determined by using the upper and lower bounds for $\tau_{\rm CO(2-1)}$ and $\tau_{\rm CO(1-0)}$ as upper and lower constant terms.

Additionally, if the calculated opacity ratios of a target deviate from LTE models, a non-LTE approach via RADEX can be explored to constrain additional absorber gas properties such as background temperature T_{BG} and molecular hydrogen density n_{H_2}.

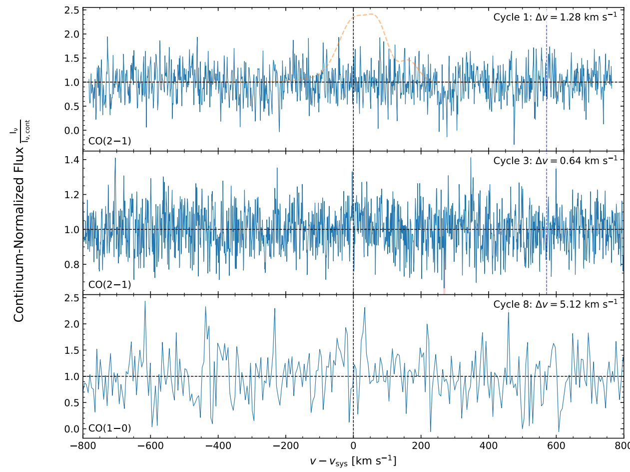
3. RESULTS

Out of the twenty targets, 10 yielded an absorption feature in at least one dataset. Results are represented by four selected non-detection cases in Figure 2 and four detection cases in Figures 3 and 4. Absorption-line measurements or upper limits for all cases, as well as associated integrated opacities and column densities for detections, are listed in Tables 3 and 4.

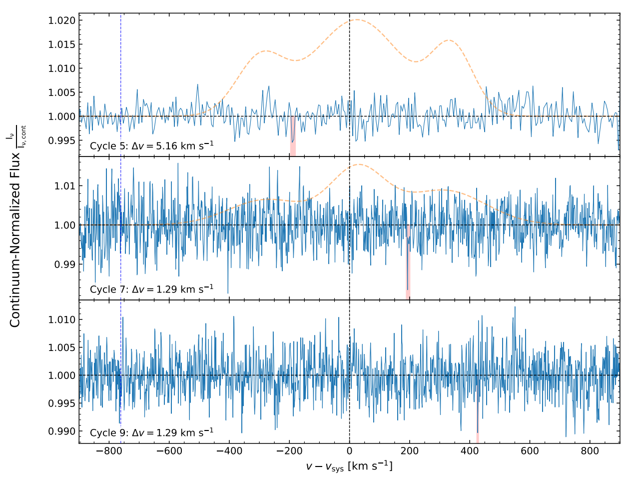
The remaining 12 spectra, as well as results for the high-z quasar calibrator target, are shown in the Appendix. When multiple data sets are present for a single target, the panels for each spectrum are organized by ALMA observation cycle number. These cycle numbers align with the years of the associated project codes, with Cycle 1 beginning in 2012, Cycle 2 in 2013, Cycle 3 in 2015, etc.



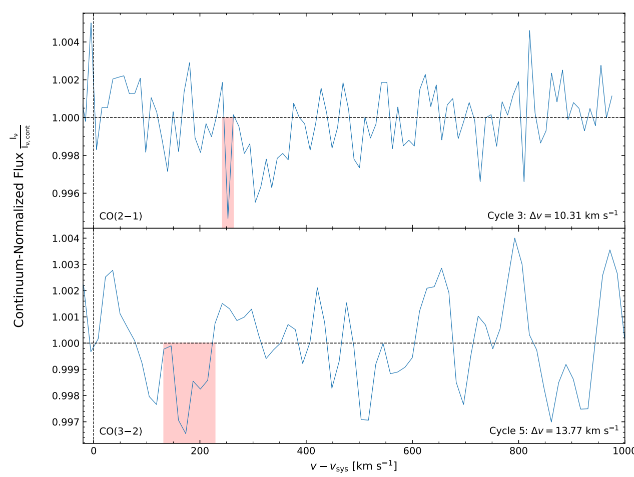
(a) ESO208-G021 CO(2–1)



(b) NGC 2110

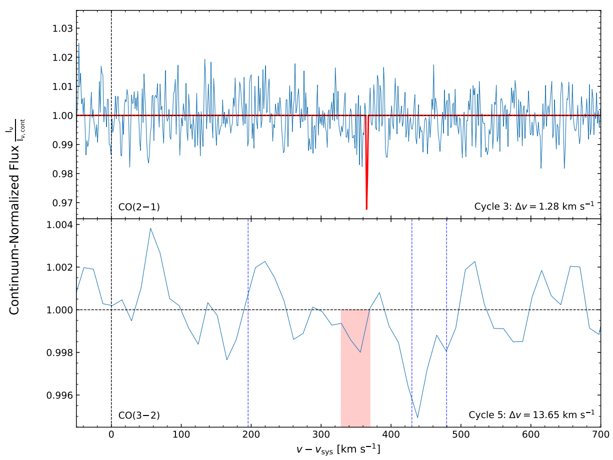


(c) NGC 315 CO(2–1)

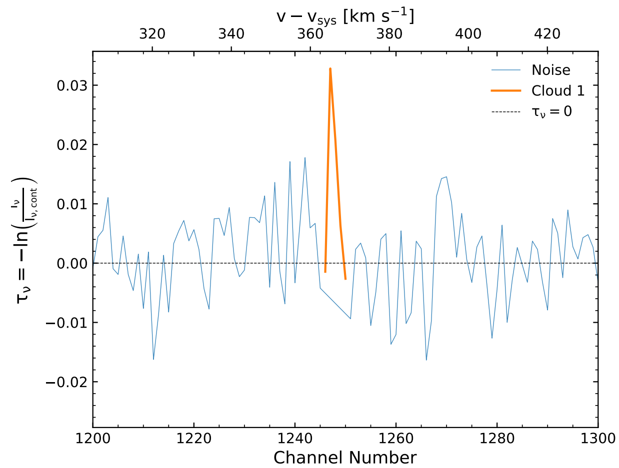


(d) NGC 6328

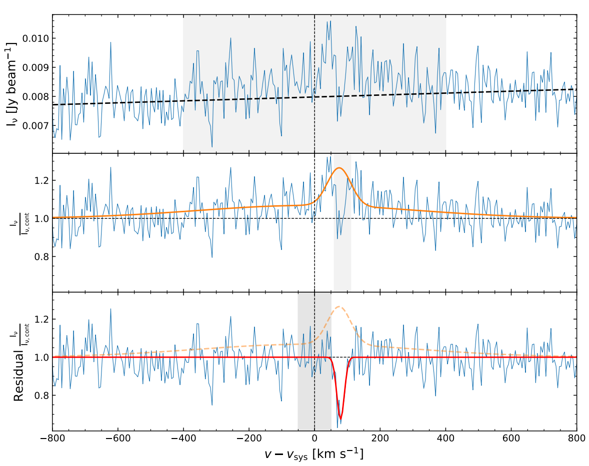
Figure 2. Continuum spectra for selected targets with non-detections of CO absorption. The vertical dashed black line marks the systemic motion of each galaxy. The horizontal black dashed curve shows the best-fit continuum level, and the yellow dashed curve shows the CO emission model. Atmospheric ozone transitions, marked by vertical dashed blue lines, may correspond to spurious positive or negative features. The velocity bin size and ALMA observing cycle corresponding to each dataset are indicated within each panel.



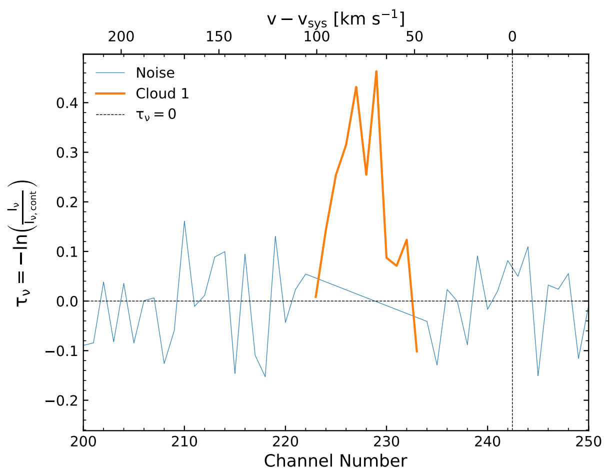
(a) IC 1459



(b) IC 1459 Optical Depth

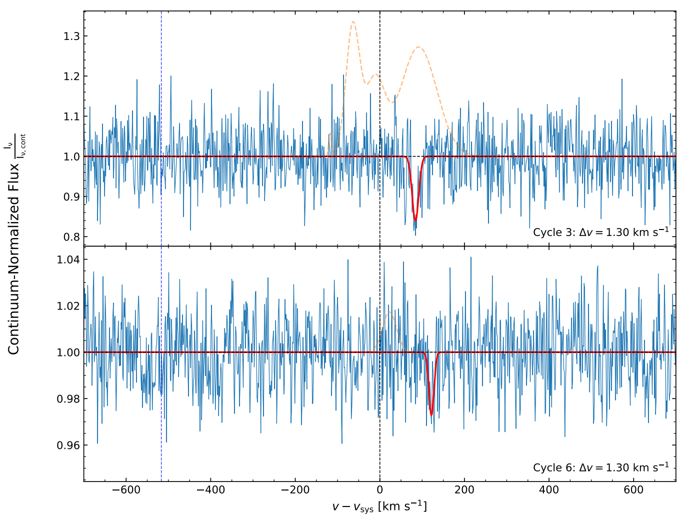


(c) NGC 1277 CO(2–1)

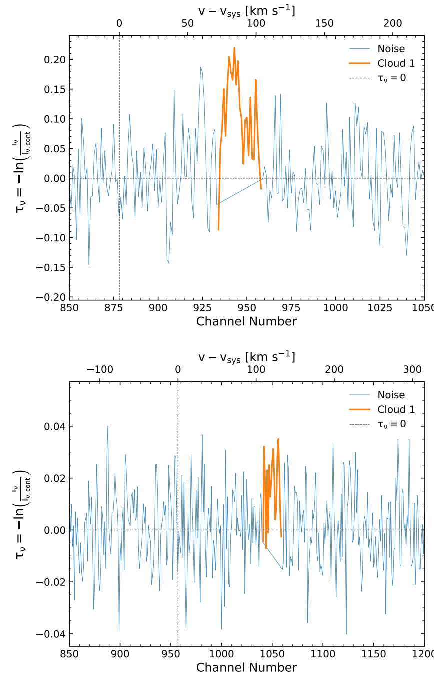


(d) NGC 1277 Optical Depth

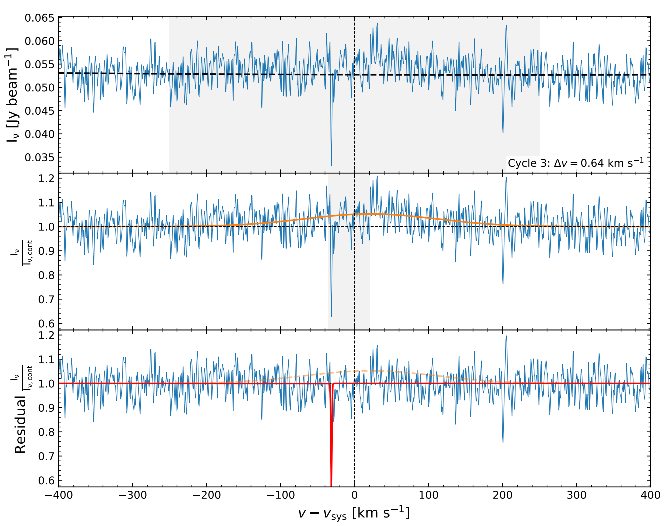
Figure 3. Continuum CO spectra for selected targets with detected CO absorption and associated optical-depth plots. The vertical dashed black line marks the systemic motion of each galaxy. The horizontal black dashed curve shows the best-fit continuum level, and the yellow dashed curve shows the CO emission model. Atmospheric ozone transitions, marked by vertical dashed blue lines, may correspond to spurious positive or negative features. The velocity bin size and ALMA observing cycle corresponding to each dataset are indicated within each panel.



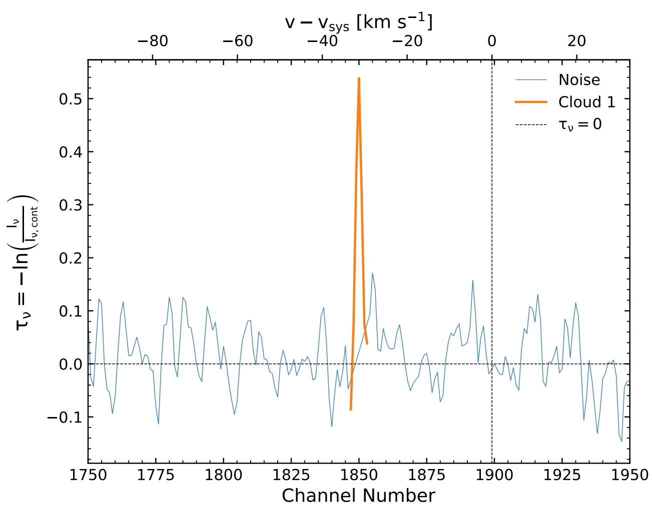
(a) NGC 3862 CO(2–1)



(b) NGC 3862 Optical Depth



(c) NGC 5084 CO(2–1)



(d) NGC 5084 Optical Depth

Figure 4. Continuum CO spectra for selected targets with CO absorption and associated optical-depth plots, continuing Figure 3.

Table 3. Upper Limits on Molecular Absorption

Source	Project code	Transition	$3\sigma_\tau \Delta v \sqrt{N_{chan}}$ (km s^{-1})	$N_{\rm CO}$ (10^{15} cm^{-2})	$N_{\rm H_2}$ (10^{18} cm^{-2})
3C 75	2017.1.01638.S	CO(3–2)	< 0.78	< 11	< 34
		CO(3–2)	< 1.45	< 20	< 63
ESO 208–G21	2017.1.00301.S	CO(2–1)	< 1.43	< 3.5	< 11
IC 1459	2017.1.01638.S	CO(3–2)	< 0.10	< 1.4	< 4.5
NGC 2110	2015.1.00419.S	CO(2–1)	< 0.53	< 1.30	< 4.06
NGC 315	2017.1.00301.S	CO(2–1)	< 0.077	< 0.19	< 0.59
	2019.1.00036.S	CO(2–1)	< 0.061	< 0.15	< 0.47
	2022.1.01040.S	CO(2–1)	< 0.031	< 0.076	< 0.24
	2015.1.00419.S	CO(2–1)	< 0.698	< 1.71	< 5.35
NGC 383	2016.1.00437.S	CO(2–1)	< 0.379	< 0.93	< 2.90
NGC 3078	2015.1.00878.S	CO(2–1)	< 0.41	< 1.0	< 3.2
NGC 3557	2015.1.00591.S	CO(1–0)	< 3.11	< 6.7	< 21
NGC 4261	2017.1.00301.S	CO(2–1)	< 0.119	< 0.29	< 0.91
	2017.1.01638.S	CO(3–2)	< 0.138	< 1.92	< 5.99
	2023.1.00842.S	CO(3–2)	< 0.101	< 1.40	< 4.38
NGC 6328	2015.1.01359.S	CO(2–1)	< 0.09	< 0.21	< 0.65
	2017.1.01638.S	CO(3–2)	< 0.19	< 2.6	< 8.0
NGC 6861	2013.1.00229.S	CO(2–1)	< 1.43	< 3.5	< 1.1
	2016.1.01135.S	CO(3–2)	< 2.8	< 7	< 2.2

Note—Upper-limit calculations for a given data set. NGC 1380 imaging 2015.1.00419.S and NGC 2110 imaging 2012.1.00474.S and 2021.1.00812.S produced targets moving between channels. The associated noise for these spectra was therefore deemed too excessive for upper-limit calculations.

Table 4. Detected Absorption Components and Associated Cloud Properties

Source	Project code	Transition	v_0 (km s^{-1})	Amplitude	σ_v (km s^{-1})	$\int \tau\, dv$ (km s^{-1})	$N_{\rm CO}$ (10^{15} cm^{-2})	$N_{\rm H_2}$ (10^{18} cm^{-2})
Hydra A	2016.1.01214.S	CO(2–1)	$-190.1^{+0.5}_{-0.5}$	$-0.580^{+0.010}_{-0.020}$	$1.8^{+0.05}_{-0.05}$	4.2 ± 0.09	10 ± 0.2	32 ± 0.7
	2016.1.01214.S	CO(2–1)	$-164.6^{+0.8}_{-1.0}$	$-0.080^{+0.010}_{-0.010}$	$5.7^{+0.6}_{-0.8}$	1.2 ± 0.11	3 ± 0.3	9 ± 0.8
	2016.1.01214.S	CO(2–1)	$-153.1^{+0.2}_{-0.8}$	$-0.210^{+0.080}_{-0.030}$	$0.5^{+0.4}_{-0.3}$	0.3 ± 0.07	1 ± 0.2	3 ± 0.6
IC 1459	2015.1.01572.S	CO(2–1)	$365.6^{+0.3}_{-0.7}$	$-0.080^{+0.050}_{-0.050}$	$0.4^{+0.5}_{-0.1}$	0.05 ± 0.02	0.1 ± 0.06	0.4 ± 0.2
IC 4296	2015.1.01572.S	CO(2–1)	$-20.8^{+3.1}_{-2.0}$	$-0.021^{+0.002}_{-0.003}$	10^{+3}_{-2}	0.4 ± 0.04	1 ± 0.1	3 ± 0.3
	2015.1.01572.S	CO(2–1)	$-2.5^{+0.3}_{-0.4}$	$-0.110^{+0.010}_{-0.005}$	$3.9^{+0.2}_{-0.3}$	1.3 ± 0.04	3 ± 0.1	10 ± 0.1
	2015.1.01572.S	CO(2–1)	$13.9^{+0.6}_{-0.6}$	$-0.030^{+0.004}_{-0.005}$	$3.3^{+0.4}_{-0.6}$	1.4 ± 0.08	3.4 ± 0.2	11 ± 0.6
NGC 0547	2017.1.00301.S	CO(2–1)	$104.0^{+1.4}_{-1.4}$	$-0.082^{+0.004}_{-0.005}$	$6.08^{+0.4}_{-0.4}$	0.9 ± 1.41	2 ± 0.2	7 ± 0.7
NGC 1052	2013.1.01225.S	CO(3–2)	$-687.6^{+4.7}_{-4.6}$	$-0.040^{+0.002}_{-0.002}$	$50^{+3.2}_{-3.2}$	5.0 ± 0.15	18 ± 0.3	57 ± 0.8
	2013.1.01225.S	CO(3–2)	$-47.8^{+3.7}_{-3.6}$	$-0.120^{+0.002}_{-0.002}$	$65^{+1.2}_{-1.2}$	20.2 ± 0.19	75 ± 0.9	233 ± 0.3
	2013.1.01225.S	CO(3–2)	$346.0^{+4.1}_{-4.1}$	$-0.060^{+0.002}_{-0.002}$	$59^{+2.4}_{-2.5}$	8.4 ± 0.17	31 ± 0.4	96 ± 1.2

Table 4 *continued on next page*

Table 4 *(continued)*

Source	Project code	Transition	v_0 (km s^{-1})	Amplitude	σ_v (km s^{-1})	$\int \tau\, dv$ (km s^{-1})	$N_{\rm CO}$ (10^{15} cm^{-2})	$N_{\rm H_2}$ (10^{18} cm^{-2})
	2015.1.00591.S	CO(1–0)	$-62.3^{+11.0}_{-10.7}$	$-0.060^{+0.004}_{-0.003}$	$38^{+2.5}_{-2.5}$	3.9 ± 0.23	15 ± 0.3	46 ± 1.0
	2015.1.00591.S	CO(1–0)	$46.5^{+12.4}_{-12.8}$	$-0.040^{+0.002}_{-0.002}$	$49^{+5.8}_{-5.2}$	7.6 ± 0.25	28 ± 0.5	88 ± 1.6
	2015.1.01290.S	CO(2–1)	$-161.5^{+5.4}_{-5.5}$	$-0.030^{+0.001}_{-0.001}$	$31^{+2.0}_{-2.0}$	3.0 ± 0.12	8 ± 0.8	26 ± 0.2
	2015.1.01290.S	CO(2–1)	$-2.9^{+5.2}_{-5.2}$	$-0.080^{+0.001}_{-0.001}$	$65^{+1.1}_{-1.1}$	12.7 ± 0.14	35 ± 0.2	108 ± 0.1
	2015.1.01290.S	CO(3–2)	$-161.0^{+15.9}_{-14.0}$	$-0.020^{+0.003}_{-0.005}$	70^{+17}_{-16}	3.8 ± 0.54	14 ± 3	44 ± 11
	2015.1.01290.S	CO(3–2)	$33.4^{+8.0}_{-7.7}$	$-0.100^{+0.010}_{-0.010}$	$38^{+5.5}_{-5.8}$	12.2 ± 0.42	45 ± 6	140 ± 19
	2015.1.01290.S	CO(3–2)	$129.4^{+9.9}_{-11.7}$	$-0.080^{+0.010}_{-0.010}$	$41^{+7.6}_{-6.5}$	7.8 ± 0.42	29 ± 5	90 ± 14
NGC 1277	2024.1.00779.S	CO(2–1)	$79.3^{+2.5}_{-2.6}$	$-0.330^{+0.040}_{-0.070}$	11^{+2}_{-2}	12 ± 1.5	28 ± 4	88 ± 11
NGC 383	2019.1.00582.S	CO(2–1)	$152.1^{+0.5}_{-0.5}$	$-0.030^{+0.004}_{-0.040}$	$0.8^{+0.2}_{-0.4}$	0.06 ± 0.02	0.2 ± 0.05	0.5 ± 0.2
NGC 3862	2015.1.00598.S	CO(2–1)	$84.1^{+1.5}_{-1.5}$	$-0.160^{+0.020}_{-0.030}$	$7.5^{+1.4}_{-1.4}$	3.2 ± 0.42	8 ± 1	24 ± 3
	2018.1.00397.S	CO(2–1)	$121.9^{+3.3}_{-1.5}$	$-0.030^{+0.004}_{-0.010}$	$5.8^{+1.6}_{-2.8}$	0.4 ± 0.08	1 ± 0.2	3 ± 0.6
NGC 4374	2013.1.00828.S	CO(2–1)	$103.8^{+0.6}_{-0.6}$	$-0.090^{+0.010}_{-0.020}$	$2.6^{+0.5}_{-0.5}$	0.6 ± 0.09	1 ± 0.2	4 ± 0.7
NGC 5084	2015.1.00598.S	CO(2–1)	$-31.4^{+0.2}_{-0.2}$	$-0.430^{+0.040}_{-0.060}$	$0.7^{+0.09}_{-0.1}$	0.9 ± 0.09	2 ± 0.2	7 ± 0.7

Note—Detected absorption components and associated cloud properties for each data set.

4. DISCUSSION

Below, I review measurements (or upper limits) and place non-detections and detections in context with results from previous surveys.

4.1. *Non-detections*

Targets where CO absorption was not detected fall into one (or more) of four sub-categories: (1) targets with a single data set, (2) targets with noisy spectral data, (3) targets with multiple data sets and no clear detections, and (4) data sets with lower resolution (i.e., higher spectral binning).

The first category can be seen in Figure 2(a) for the target ESO 208–G021 and its CO(2–1) observation from Cycle 5. The three panels for the associated spectra showcase the original peak-intensity spectrum on the top, the normalized continuum spectrum in the middle, and the emission-subtracted spectrum on the bottom. The continuum itself is represented by a first-degree polynomial fit that I applied outside of the gray shaded emission region, as seen in the top panel. Another gray shaded region can be seen within the emission region of the middle panel. This region was masked when I applied the emission fitting model. If I had instead chosen not to apply the mask, then the Gaussian model would have had a lower amplitude. If absorption is present within emission, then it is important for the emission model to be applied correctly so as not to over- or undersaturate the absorption dip. Within the spectra of ESO 208–G021, there may be an absorption feature that begins around $v \approx v_{\rm sys}$ and ends at about 10 km s^{-1}. One can choose to model this feature as absorption; however, I decided against it due to its rather low prominence compared to some of the adjacent features, particularly at $v \approx -250$ km s^{-1} and 300 km s^{-1}. The absence of another data set for comparison also led me to interpret the feature as noise rather than as a cloud within the Earth's line of sight. Follow-up observations of this galaxy, with coverage of CO(2–1) and CO(1–0), may support or refute this claim.

Low S/N often prevents the identification of absorption features, regardless of careful self-calibration or imaging procedures. This can be seen in Figure 2(b) for the target NGC 2110. Three data sets were available for this target, with two CO(2–1) transitions and one CO(1–0) transition. Emission in the Cycle 1 spectrum was subtracted out. While examining the residual spectra, there may be an extended feature between $v \approx 250$–$300\ \mathrm{km\,s^{-1}}$ and a stronger one at $v \approx 475\ \mathrm{km\,s^{-1}}$. However, the noisy data, the extension of the latter feature into negative intensities, and the lack of analogous features in the less noisy Cycle 3 data minimize any corroborating evidence. Within the Cycle 8 CO(1–0) spectrum, the resolution of the data itself is also about five times lower than that of the preceding two, while the noise is comparable to that of the Cycle 1 data, preventing additional analysis. Since noisy spectral data may have unphysical τ_{CO} values, I took the upper limits in Table 3 from the higher-S/N Cycle 3 data.

Data sets with similarly high S/N allow for a detailed examination of faint CO absorption. Within Figure 2(c), three CO(2–1) data sets for NGC 315 are shown. In this case, two finer-resolution spectra from Cycles 7 and 9 are compared to a coarser spectrum from Cycle 5. In the lower-angular-resolution Cycle 5 and 7 data, I removed underlying CO emission blurred into the central pixel with a triple-Gaussian fit. Unfortunately, none of these three spectra show a clear absorption feature. The red highlighted region in the middle panel could be a feature; however, additional prominent features within the data set suggest that it is more likely noise than a real detection. It can also be said that the emission model for the Cycle 5 data may have overfitted the spectra, as there is a clear drop in intensity when moving between the outside noise region and the inside emission region. Multiple high-S/N data sets give confidence that there is no obvious detection.

The final case, shown in Figure 2(d) for the target NGC 6328, is data with lower spectral resolution. Data sets for CO(2–1) from Cycle 3 and CO(3–2) from Cycle 5 both show no obvious detection. The binning of both spectra is also relatively high, at $\sim 11\ \mathrm{km\,s^{-1}}$. This implies that any faint feature with $\sigma_{\mathrm{v}} < \Delta v$ would instead be averaged out with adjacent velocity channels and ultimately not be observed. Since our interest is to uncover fainter features with low σ_{v}, a data set with high Δv (low resolution) will not aid in our census of these fainter lines. However, upper-limit detections can still be made since the noise level is relatively low, as can be seen by the very low spread of 0.003 in the normalized intensity.

4.2. *Detections*

Among the targets where CO absorption is detected, four interesting cases are shown that depict the wide range of absorber types in these target galaxies.

The first case is one in which a faint absorption feature is present in one data set, but not in the adjacent one for a different CO transition. Figure 3(a) showcases two spectra, Cycle 3 CO(2–1) and Cycle 5 CO(3–2), for IC 1459. In the Cycle 3 data set, there exists a very faint feature at $v \approx 365\,\mathrm{km\,s^{-1}}$, whose $\int \tau\, dv = 0.05 \pm 0.02\ \mathrm{km\,s^{-1}}$, corresponding to an optically thin cloud with a full width at half maximum (FWHM) that matches, or is smaller than, the velocity bin size. However, this feature does not appear at all in the CO(3–2) Cycle 5 data due to the 10× lower spectral resolution. A prominent dip at a higher velocity, around $v \approx 450\,\mathrm{km\,s^{-1}}$, coincides with the rest frequencies of two atmospheric ozone lines, but nothing appears near the centroid velocity of the faint CO absorber. Potentially, the reason why the faint feature is not seen in the Cycle 5 spectrum is that the resolution is too coarse to resolve it. With $\Delta v = 13.64\,\mathrm{km\,s^{-1}}$, this is about 27 times the σ_{v} of the detected feature, meaning that the feature was likely averaged out with adjacent velocity channels.

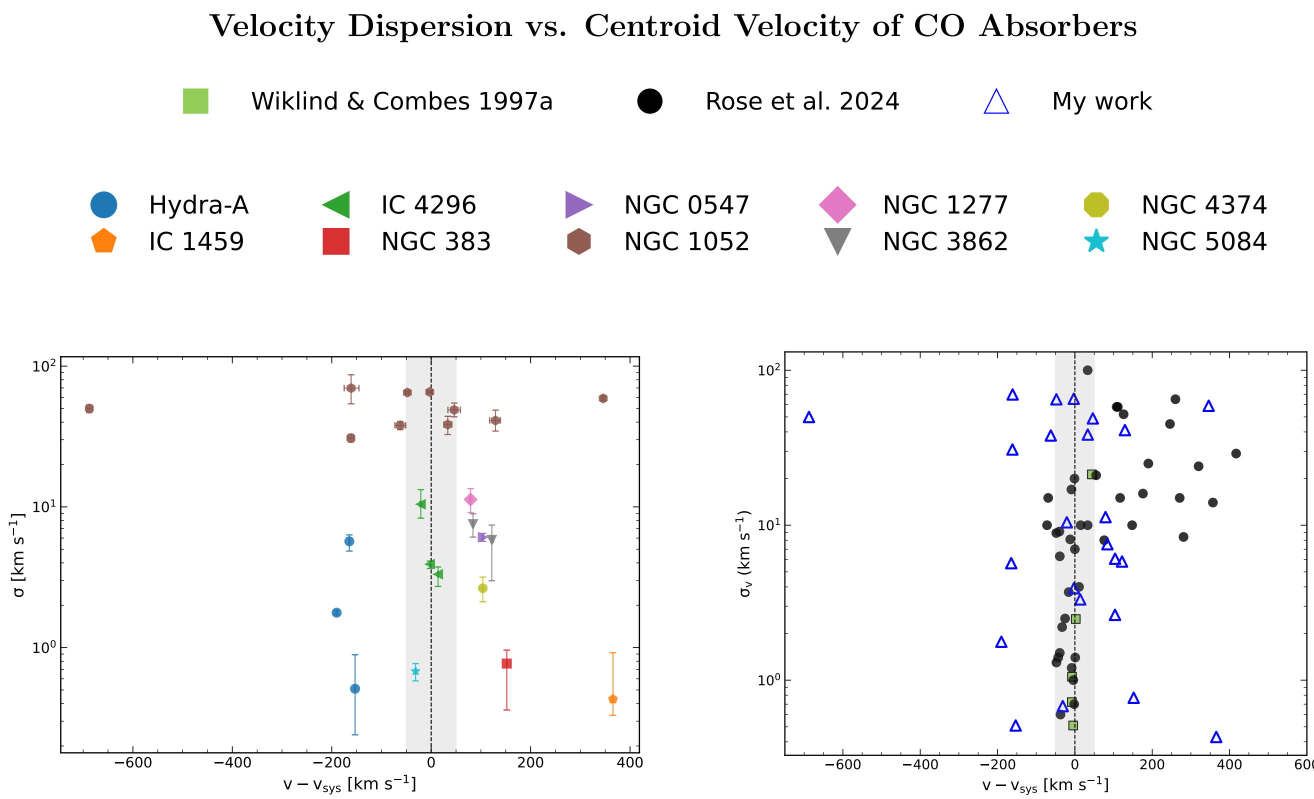
Regardless, an upper limit can be obtained by regarding the associated dip in the Cycle 5 spectrum as a mock cloud. For the Cycle 5 CO(3–2) data, I estimate an upper limit of $\sigma_\tau \Delta v \sqrt{N_{\rm chan}} = 0.10\,{\rm km\,s^{-1}}$. From the Cycle 3 data, the absorbing cloud has an $N_{\rm H_2} = 0.4 \pm 0.2 \times 10^{18}\,{\rm cm^{-2}}$.

The next case is one in which an absorption feature is embedded within emission. Figure 3(c) shows a CO(2–1) observation of NGC 1277 from Cycle 11. Emission blended into the peak continuum spectrum is clearly present between $v - v_{\rm sys} = \pm 400$ km s^{-1}. I fit and remove the emission using a single-Gaussian model after masking the absorption-line channels. Preliminary masking of too many channels led to a higher amplitude for the Gaussian emission term, which then produced a deeper absorption feature within the residual spectrum. As this process is somewhat subjective, the choice of masking contributes to a systematic error in the absorption-line parameters. However, this error analysis will be performed in future work. Regardless of the masked region, the centroid velocity remains relatively unchanged, with $v - v_{\rm sys} = 79.3^{+2.5}_{-2.6}$ km s^{-1}, suggesting infalling gas or an extreme outlier among disk-like clouds. The feature in NGC 1277 is tightly constrained, with $\sigma_{\rm v} = 11 \pm 2$ km s^{-1} and $\int \tau\, dv = 12 \pm 1.5$ km s^{-1}. The velocity dispersion and integrated opacity are consistent with either a disk-like or infalling optically thick cloud. Additional observations of NGC 1277 may aid in the further confirmation of this feature and its potential evolution with time.

A time evolution may be seen in the two CO(2–1) spectra of NGC 3862. Two different data sets from Cycle 3 and Cycle 6, with the same spectral resolution, show a similar dip around $v - v_{\rm sys} \approx 100$ km s^{-1} in Figure 4(a). However, the centroid velocities for the noisier Cycle 3 and Cycle 6 features, $v - v_{\rm sys} = 84.1 \pm 1.5$ km s^{-1} and $v - v_{\rm sys} = 121.9^{+3.3}_{-1.5}$ km s^{-1}, respectively, do not coincide. Their respective dispersions are $\sigma_{\rm v} = 7.5 \pm 1.4$ km s^{-1} and $\sigma_{\rm v} = 5.8^{+1.6}_{-2.8}$ km s^{-1}, and their integrated opacities are $\int \tau\, dv = 3.2 \pm 0.42$ km s^{-1} and $\int \tau\, dv = 0.4 \pm 0.08$ km s^{-1}. Given the lower noise, the Cycle 6 data would not fail to detect the feature seen in Cycle 3. The time difference between the two observations is about 3 years, perhaps suggesting either an optically thin cloud that increased its infall velocity by about 40 km s^{-1} over the span of three years, or a line of sight through a distinct cloud.

The final case can be seen in Figure 4(c) for the target NGC 5084. A single CO(2–1) data set from Cycle 3, with a resolution of $\Delta v = 0.64$ km s^{-1}, highlights an extremely faint feature analogous to that detected in front of IC 1459. There is emission present in the spectrum between $v = \pm 250$ km s^{-1}, which I masked when applying the continuum fit. Another masked region within the emission shows clear dips between -50 km s^{-1} and 25 km s^{-1}, which I also masked when applying the emission model. The overall emission Gaussian may be underfitting the emission feature, as can be seen from the prominence of the two positive peaks around 35 km s^{-1}. However, it works well enough to bring the emission to the continuum level while preserving embedded absorption. The modeled feature itself has $v_0 = -31.4 \pm 0.2$ km s^{-1}, $\sigma_{\rm v} = 0.7 \pm 0.1$ km s^{-1}, and $\int \tau\, dv = 0.9 \pm 0.09$ km s^{-1}. There appears to be another absorption feature at ≈ 200 km s^{-1}; however, this may be due either to observational frequency-range constraints, absorption from another molecular species, or contamination from an atmospheric transition line. The main deep, faint feature, however, is prominent relative to the surrounding noise and is tightly constrained by the Monte Carlo resampling, thereby suggesting the observation of a faint, optically thin cloud in circular motion around the galactic nucleus of NGC 5084.

4.3. *Absorber Components Plotted*



(a) Our Absorption Detections

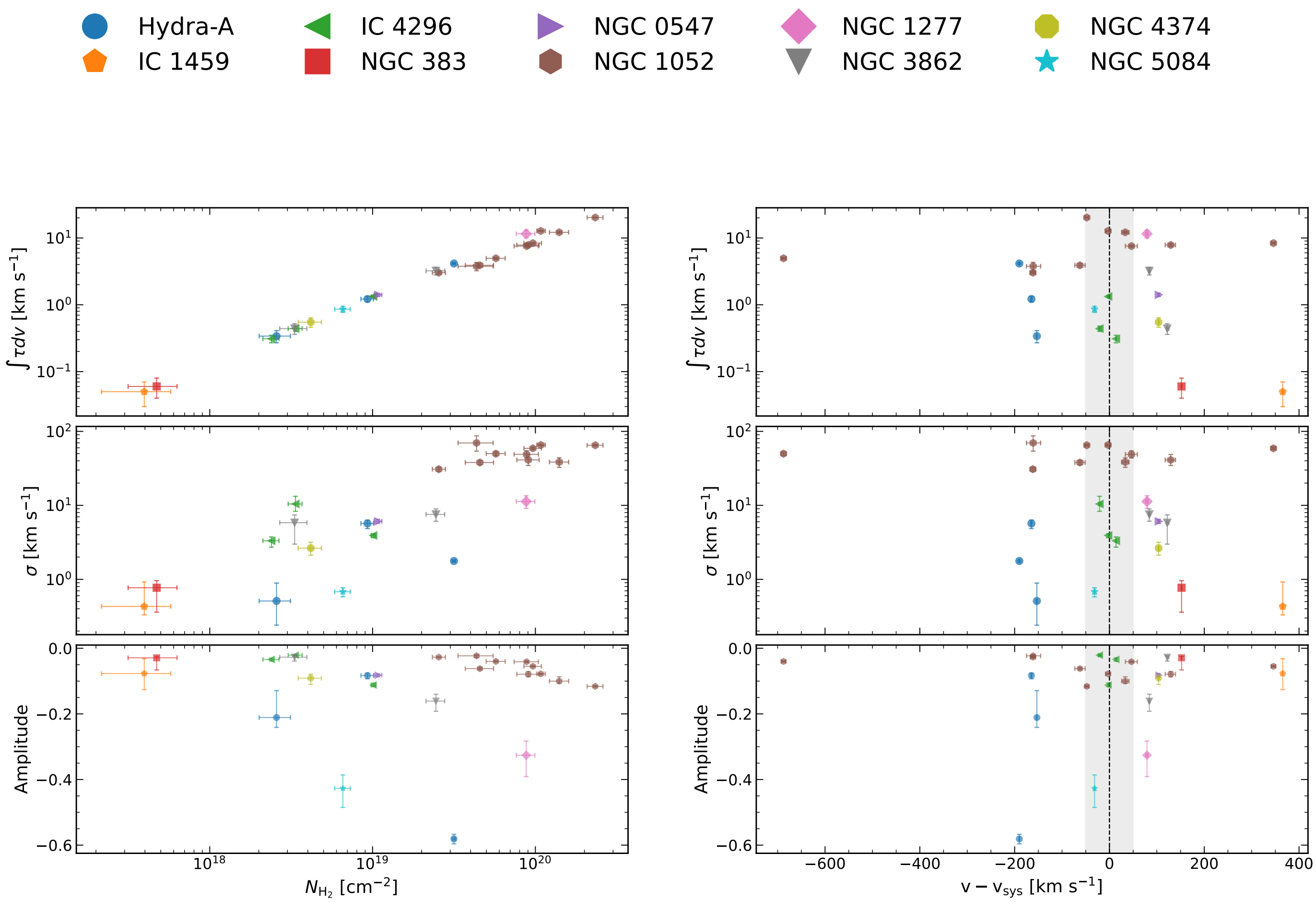
(b) Combined With Previous Absorption Studies

Figure 5. Comparison between the velocity dispersion and centroid velocity of the detected absorbers in this study (a), along with previously detected absorbers from Rose et al. (2024) and Wiklind & Combes (1997a) (b), with the gray region being the typical velocity range for disk-like clouds moving near the systemic motion of their host galaxies.

Based on the distribution of detected absorbers from Figure 5(a), three populations of molecular gas clouds exist regardless of how optically thick the clouds are, namely inflows, outflows, and gaseous disk material. The most optically thick clouds in our sample are seen in the broad spectral features ($\sigma_v \gtrsim 30$ km s^{-1}) of NGC 1052, while optically thin clouds with $\sigma_v < 10$ km s^{-1} are detected in the remaining systems. In comparison to the detections reported by Wiklind and Combes (1997a) and Rose et al. (2024) in Figure 5(b), clouds with low velocity dispersion tended to represent a population with velocities similar to the systemic motion of their host galaxies. However, our findings suggest that there does exist a population of optically thin clouds that are inflowing toward the central galactic region rather than being in circular motion. These faint absorbers are found in IC 1459 and NGC 383. The absorbers of NGC 0547, NGC 3862, and NGC 4374 all have lower σ_v in comparison to the other outflowing clouds found in Rose et al. (2024).

The data from Hydra A may also suggest a population of diffuse outflows; however, the centroid velocities can change substantially based on the $z_{\rm sys}$ value used. Since Dr. Boizelle cleans all observational data using the barycentric reference frame, I found it appropriate to use the heliocentric redshift for $z_{\rm sys}$ from NED for each target. The difference between the barycentric and heliocentric frames is negligible, as the former uses the solar system's center of mass and the latter the frame of the Sun. With this in mind, Hydra A was also studied by Rose et al. (2019, 2020). They calculated their own heliocentric redshift of $z_{\rm sys} = 0.0543519$ in comparison to $z_{\rm sys} = 0.054878$ from NED. This results in about a 150 $\rm km\,s^{-1}$ difference in centroid velocities. If I had instead used their quoted redshift value, then the Hydra A centroids would fall within the $\pm\,50$ $\rm km\,s^{-1}$ region and would be more consistent with the expected cloud motion for Hydra A's disk. This redshift disparity can also be extended to NGC 1052, as Rose et al. (2024) used the de Vaucouleurs et al. (1991) value of $z_{\rm sys} = 0.00498$, whereas I used $z_{\rm sys} = 0.005204$ from Koss et al. (2022), resulting in a 66 $\rm km\,s^{-1}$ centroid-velocity difference. This would not affect the locations of the detected absorbers in NGC 1052 as much as it would for Hydra A. Regardless, there should be some universal redshift reference frame used when analyzing spectra for CO absorption, as different values can cause certain cloud populations either to be missed or to be artificially created.

Integrated Opacity, Velocity Dispersion, Amplitude vs. Column Density, Centroid Velocity



(a) Column Density Relations

(b) Centroid Velocity Relations

Figure 6. The integrated opacity, velocity dispersion, and amplitude of detected absorbers are plotted against corresponding column densities (a) and centroid velocities (b). The gray region is the typical velocity range for disk-like clouds moving near the systemic motion of their host galaxies.

From the calculable quantities obtained, as well as the best-fit Gaussian parameters, I attempted to uncover potential relations by plotting $\int \tau\, dv$, σ_{v}, and the amplitude against both $N_{\mathrm{H_2}}$ and $v - v_{\mathrm{sys}}$. These plots can be seen in Figure 6. In Figure 6(a), the top panel shows an exact linear relationship between $\int \tau\, dv$ and $N_{\mathrm{H_2}}$. This is to be expected, as the integrated opacity scales linearly with column density, as seen in Equation 9. In the middle panel, an interesting relation between σ_{v} and $N_{\mathrm{H_2}}$ is observed, as they appear to scale linearly. This implies that broader features have more gas in Earth's line of sight, which is also what one would expect. Likewise, narrower features translate to less gas along that line of sight. The bottom panel shows no direct correlation between the amplitude of the absorption dip and its associated $N_{\mathrm{H_2}}$. A straight line can almost be drawn from an amplitude value of -0.1 to encompass varying column-density values. Among the narrow features, a lower amplitude tends to correspond to more line-of-sight gas than a fainter amplitude. This can be seen in IC 1459 and NGC 383, as both have amplitudes > -0.1, whereas Hydra A and NGC 5084 have absorbers of similar $\sigma_{\rm v}$ but even lower amplitudes.

CONCLUSION

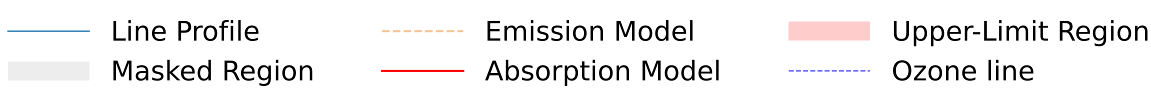
Out of a sample of twenty nearby elliptical and lenticular galaxies, half were observed to have a low-J CO rotational absorption line. Faint, narrow lines from IC 1459 and NGC 383 ($\sigma_{\rm v} < 1$ km s^{-1}) point toward a population of diffuse, faint, optically thin inflowing molecular gas. Slightly broader lines from NGC 0547, NGC 3862, and NGC 4374, with $\sigma_{\rm v}$ between 2 and 8 km s^{-1}, can also belong to this population of gas. The narrower lines from IC 1459 and NGC 383 are the result of high-spectral-resolution, low-velocity-binning observations with $\Delta v \lesssim 2$ km s^{-1}. Higher-velocity-binning observations of similar targets may miss this population of clouds entirely. Similar absorbing clouds from Hydra A may suggest another population of faint, diffuse outflows; however, additional discussion regarding exact $z_{\rm sys}$ values is required to confirm or refute these results.

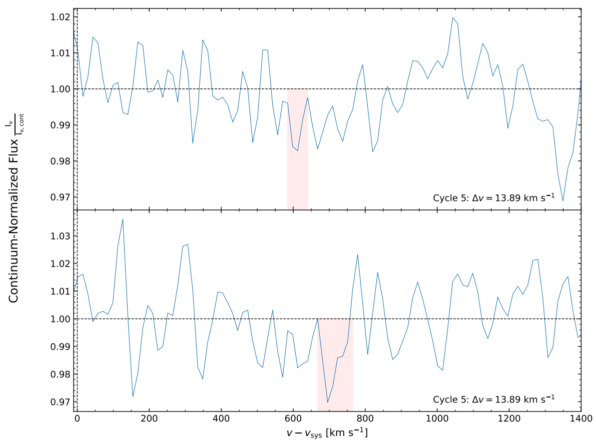
ACKNOWLEDGMENTS

I would like to especially thank Dr. Boizelle for guiding me through the course of this research over the past two semesters. I would also like to thank Dr. Baker for his invaluable suggestions on submittable documentation as well.

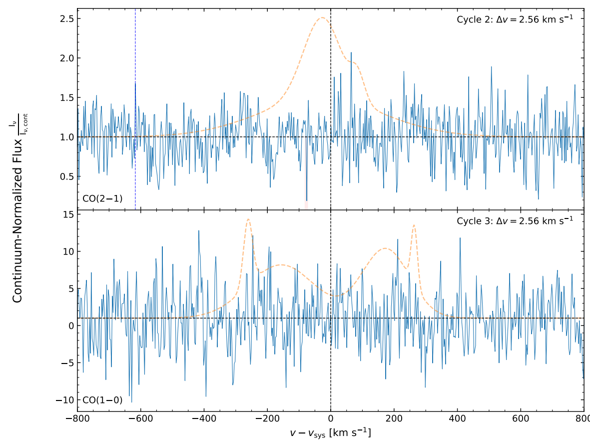
APPENDIX

The first section of the appendix showcases the CO spectral lines for the remaining 12 targets. They are organized as follows: Figure 7 shows the 6 remaining non-detection cases, Figure 8 shows an additional 3 detection cases, and Figure 9 shows an additional 2 detection cases, while NGC 1052 has its own dedicated section multiple CO transitions. At the end, the spectral lines for J1427-4206 is shown.

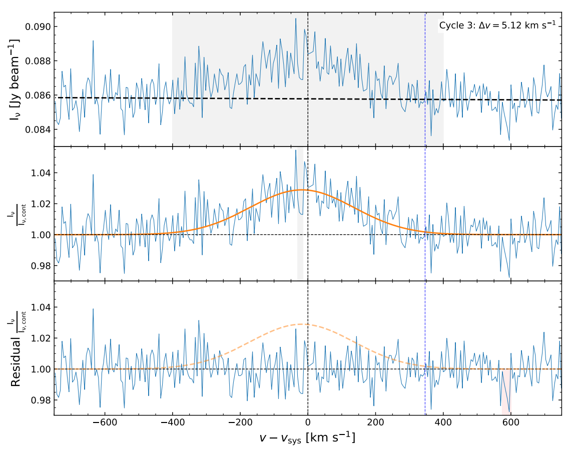




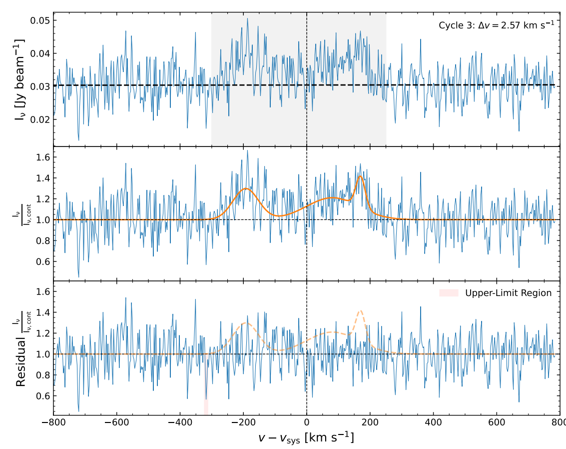
(a) 3C 75 CO(3–2) Spectral Lines



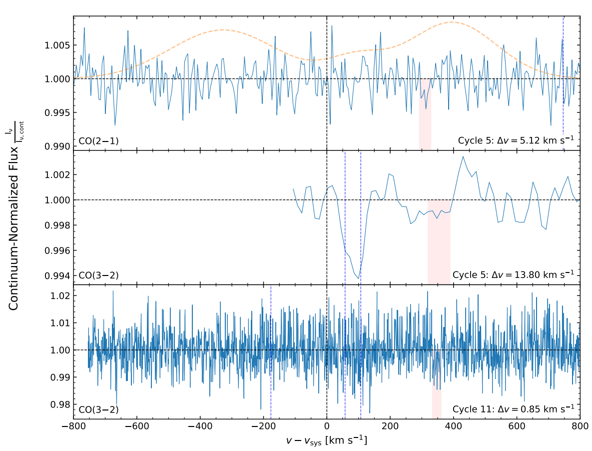
(b) NGC 1380 CO Spectral Lines



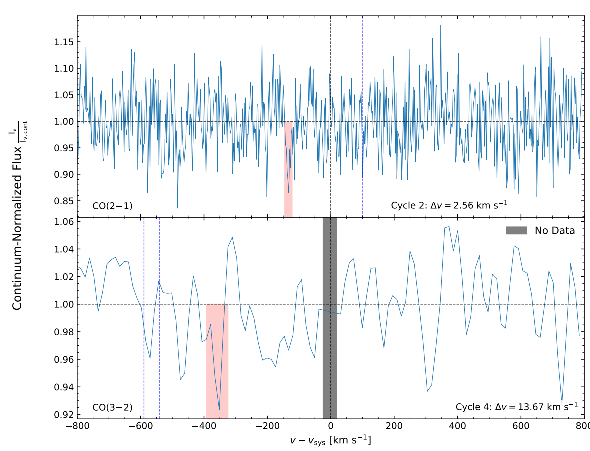
(c) NGC 3078 CO(2–1)



(d) NGC 3557 CO(1–0) Spectral Line

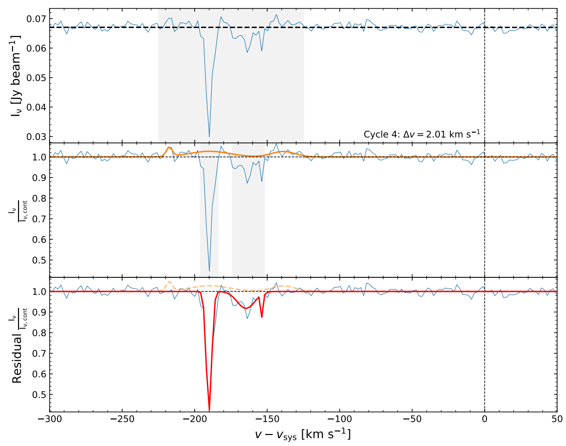


(e) NGC 4261 Spectral Lines

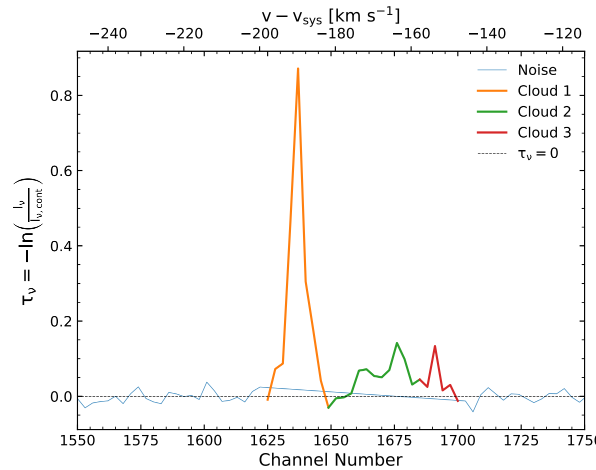


(f) NGC 6861 CO Spectral Lines

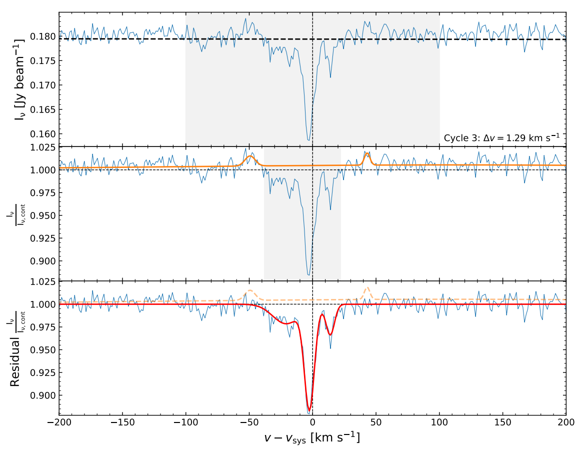
Figure 7. Remaining 6 non-detection cases.



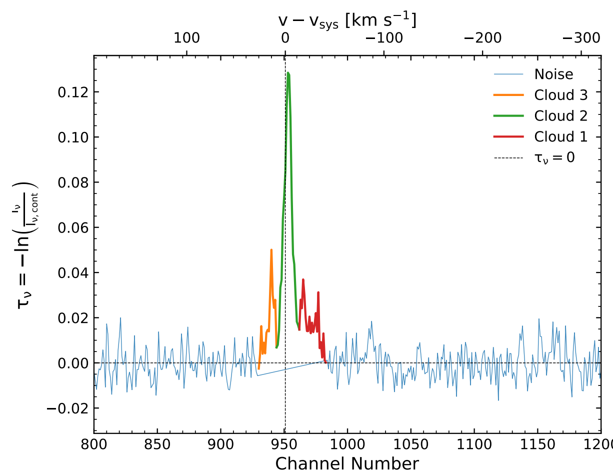
(a) Hydra A CO(2–1) Spectral Line



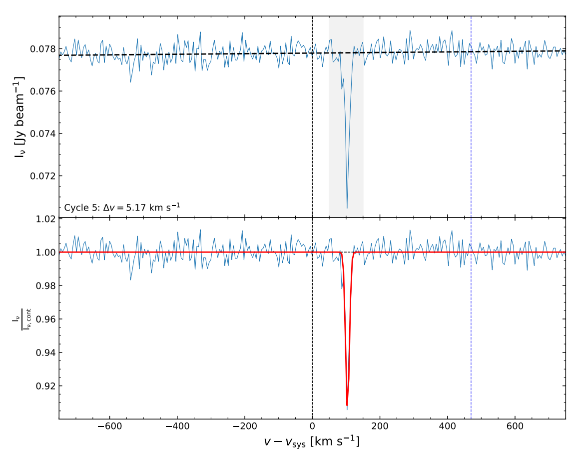
(b) Hydra A Optical Depth



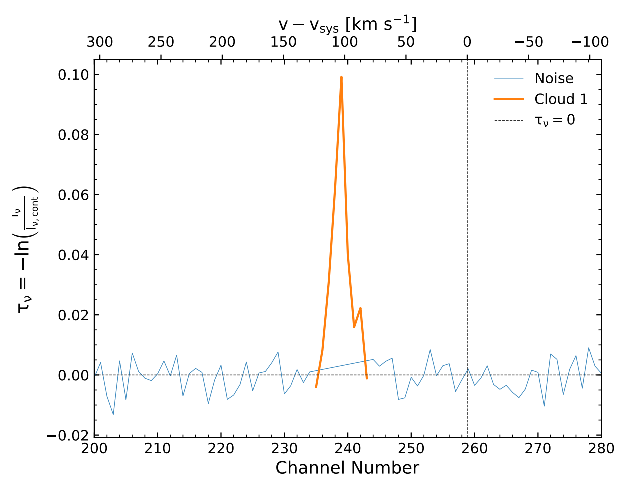
(c) IC 4296 CO(2–1) Spectral Line



(d) IC 4296 Optical Depth

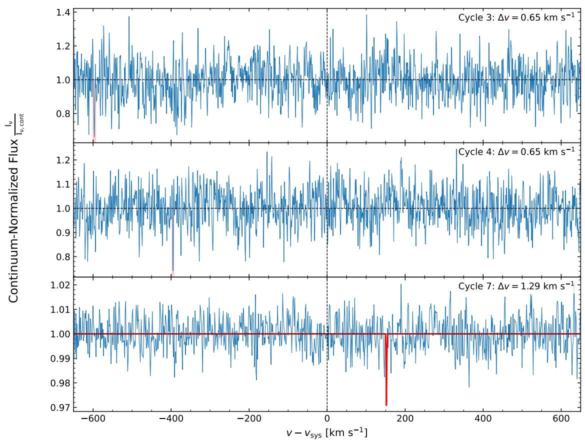


(e) NGC 0547 CO(2–1) Spectral Line

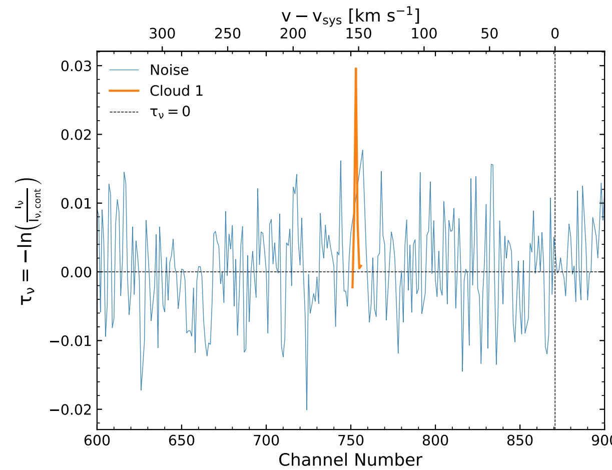


(f) NGC 0547 Optical Depth

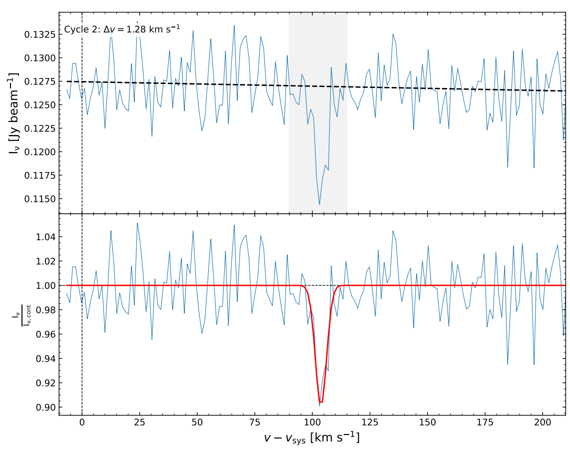
Figure 8. Additional 3 detection cases.



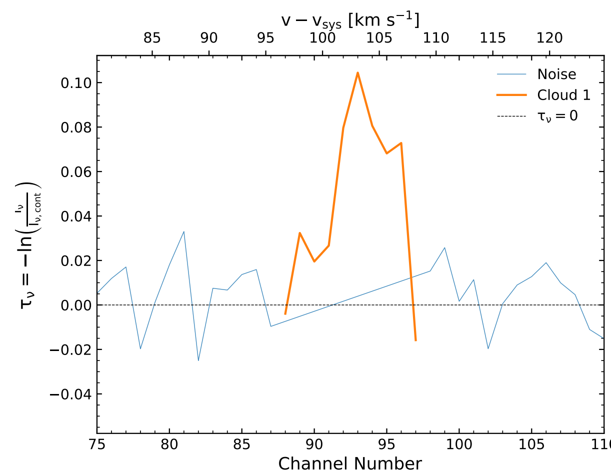
(a) NGC 383 CO(2–1) Spectral Lines



(b) NGC 383 Cycle 7 Optical Depth

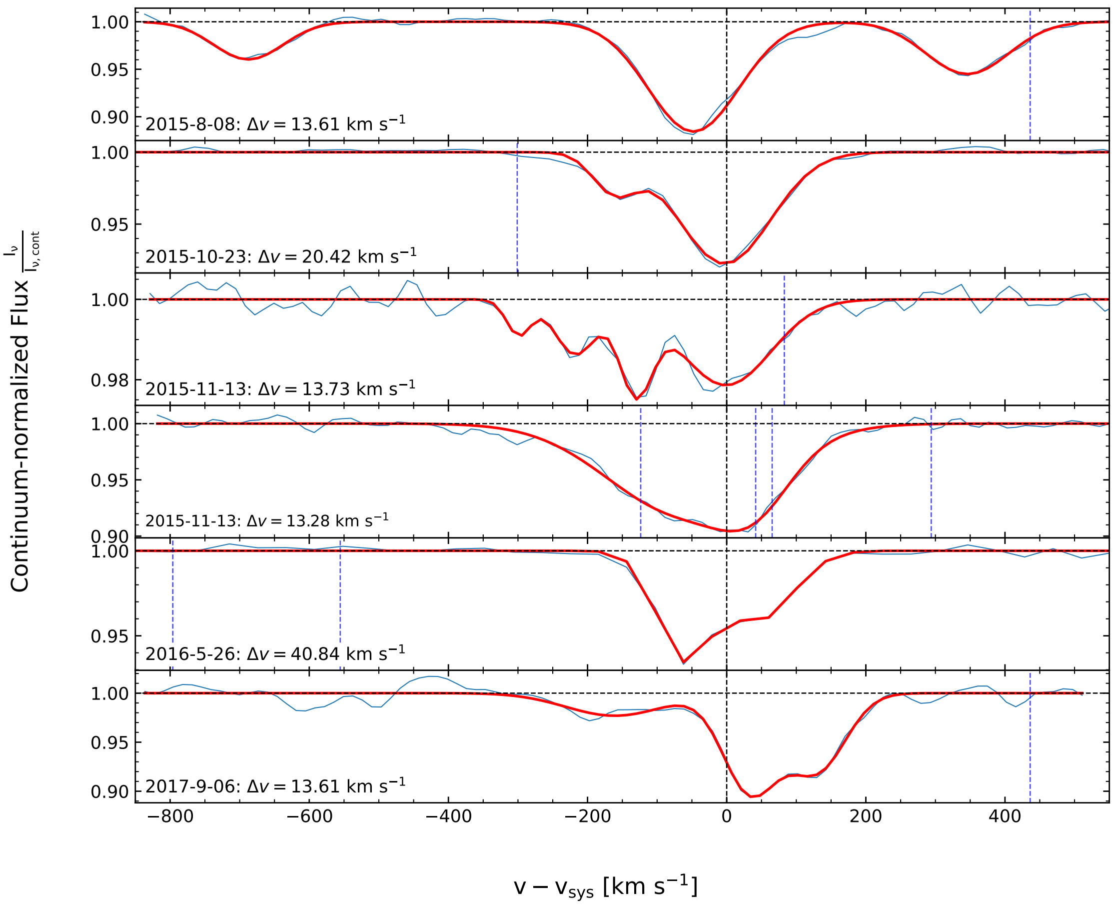


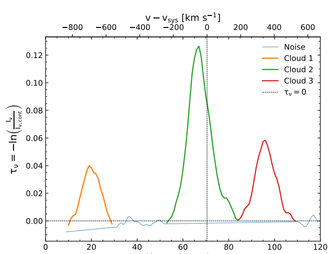
(c) NGC 4374 CO(2–1) Spectral Line



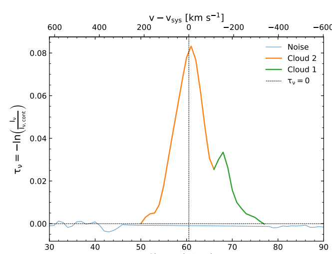
(d) NGC 4374 Optical Depth

Figure 9. Additional 2 detection cases.

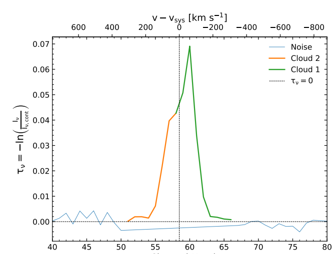




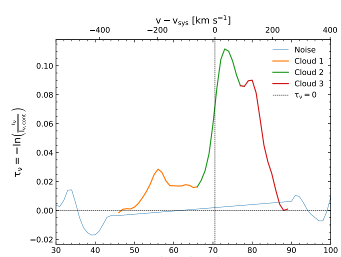
(a) 2015-08-08



(b) 2015-10-23



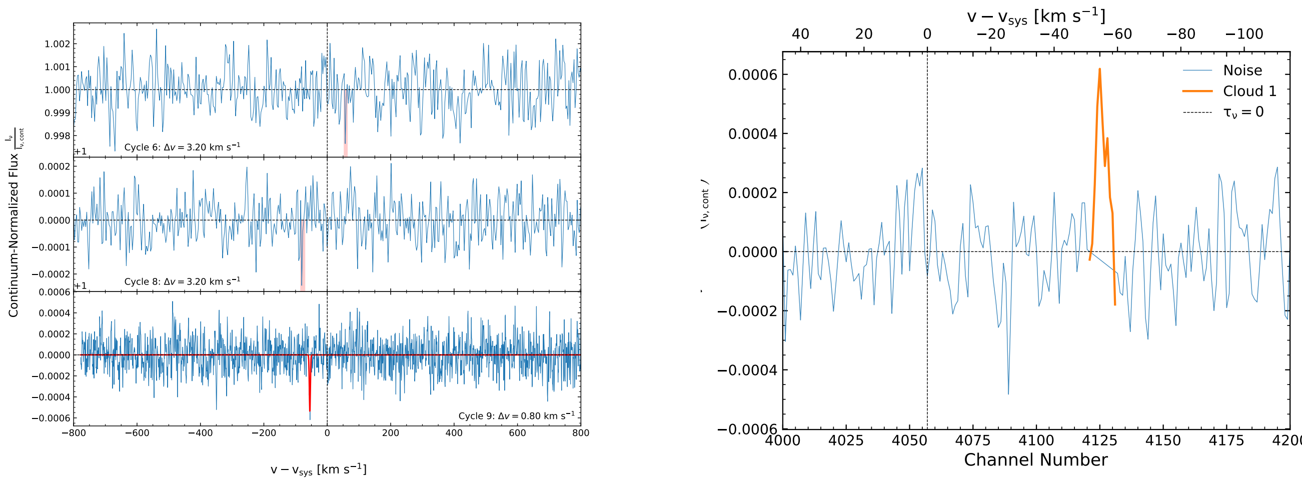
(c) 2016-05-26



(d) 2017-09-06

Figure 10. NGC 1052 spectral lines and integrated optical depth plots for the four CO absorptionlines, ordered chronologically by observation date.

J1427–4206



(a) J1427–4206 CO(2–1) Spectral Lines

(b) Cycle 9 Optical Depth

Figure 11. J1427–4206 (High-z quasar calibrator target) spectral lines and integrated optical depth plots.

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